

Koopman Operator Analysis of Nonlinear Dynamical Systems

B.Tech DC Presentation

April 20, 2026

Nonlinear dynamical systems appear in many physical systems:

- Planetary motion
- Fluid dynamics
- Robotics
- Control systems

Traditional analysis is difficult — nonlinear systems cannot usually be solved analytically.

Koopman operator theory provides a way to analyze nonlinear systems using linear techniques.

Linear vs Nonlinear Systems

A **linear system** follows two laws:

- ① **Homogeneity:** $T(ax) = aT(x)$
- ② **Additivity:** $T(x_1 + x_2) = T(x_1) + T(x_2)$

Linear systems have closed-form solutions:

$$\dot{x} = Ax \Rightarrow x(t) = e^{At}x_0$$

Nonlinear systems (e.g. $\sin \theta$ terms) generally have **no closed-form solution** $\Rightarrow$ new tools required.

Koopman Operator Concept

Instead of studying the state x , study **functions of the state** called **observables**:

$$g : \mathbb{R}^n \rightarrow \mathbb{R}$$

Examples: $g(x) = x_1$, $g(x) = x_1^2$, $g(x) = \sin(x_1)$

For a discrete system $x_{k+1} = F(x_k)$, the **Koopman operator** $\mathcal{K}$ is:

$$(\mathcal{K}g)(x) = g(F(x)) = g(x_{k+1})$$

Key property: Even if F is nonlinear, $\mathcal{K}$ is **always linear**:

$$\mathcal{K}(ag_1 + bg_2) = a\mathcal{K}g_1 + b\mathcal{K}g_2$$

Why Koopman? The Core Idea

	State space	Koopman space
Dimension	$\mathbb{R}^n$ (finite)	$\mathcal{G}$ (infinite)
Dynamics	Nonlinear F	Linear $\mathcal{K}$
Tools available	Limited	Full linear algebra

The idea: Make the system linear in *function space*, not state space:

$$\underbrace{\dot{x} = f(x)}_{\text{nonlinear in } \mathbb{R}^n} \longrightarrow \underbrace{\dot{\Psi} = \mathcal{K}\Psi}_{\text{linear in } \mathcal{G}}$$

In practice: approximate $\mathcal{K}$ on a finite dictionary $\Psi(x) \in \mathbb{R}^N$ using data.

How diagonalization arises in the Koopman operator framework.

A discrete-time dynamical system is defined as:

$$x_{n+1} = f(x_n) \quad (1)$$

Repeated application gives:

$$x_m = f^{(m)}(x_0) \quad (2)$$

Instead of tracking the state x , we consider functions of the state called **observables**:

$$\varphi : X \rightarrow \mathbb{R} \quad (3)$$

Examples include $\varphi(x) = x, x^2, \sin x$.

The Koopman operator $\mathcal{K}$ acts on observables as:

$$(\mathcal{K}\varphi)(x) = \varphi(f(x)) \quad (4)$$

Applying multiple times:

$$\mathcal{K}^m \varphi(x) = \varphi(f^{(m)}(x)) \quad (5)$$

An eigenfunction ϕ_λ satisfies:

$$\mathcal{K}\phi_\lambda = \lambda\phi_\lambda \quad (6)$$

This implies:

$$\phi_\lambda(f(x)) = \lambda\phi_\lambda(x) \quad (7)$$

The observable scales by a constant factor λ after one step of dynamics.
Applying $\mathcal{K}$ repeatedly:

$$\mathcal{K}^m\phi_\lambda = \lambda^m\phi_\lambda \quad (8)$$

This shows exponential-type evolution.

If we can find a basis of eigenfunctions, any observable can be expanded as:

$$\varphi = \sum_k v_k \phi_{\lambda_k} \quad (9)$$

Applying $\mathcal{K}^m$:

$$\mathcal{K}^m\varphi = \mathcal{K}^m\left(\sum_k v_k \phi_{\lambda_k}\right) = \sum_k v_k \mathcal{K}^m\phi_{\lambda_k} = \sum_k v_k \lambda_k^m \phi_{\lambda_k} \quad (10)$$

For a matrix A :

$$Av = \lambda v \quad (11)$$

Then:

$$A^m v = \lambda^m v \quad (12)$$

And for any vector:

$$x = \sum_k c_k v_k \quad (13)$$

We get:

$$A^m x = \sum_k c_k \lambda_k^m v_k \quad (14)$$

Koopman operator provides a linear representation of nonlinear dynamics in function space.

- Nonlinear system in state space
- Linear evolution in observable space
- Eigenfunctions behave like independent modes

Diagonalizing the Koopman operator simplifies the study of complex nonlinear systems by transforming them into independent exponentially evolving components.

JUICE Analogy — Pendulum

Goal: Use JUICE mission data to detect subsurface oceans on Jupiter's moons.

The pendulum analogy:

- System: mass on a string
- Forcing: periodic external push
- Observable: swing amplitude
- Inference: amplitude reveals stiffness (system properties)

For icy moons (Europa, Ganymede, Callisto):

- Forcing: Jupiter's periodic gravitational pull
- Observable: tidal bulge amplitude (~ 1 m vs ~ 30 m)
- Inference: large bulge $\Rightarrow$ subsurface ocean present

Nonlinear Pendulum Model

The nonlinear pendulum equation:

$$\ddot{\theta} + \frac{g}{L} \sin(\theta) = 0$$

State variables: $x_1 = \theta$, $x_2 = \dot{\theta}$

First-order form:

$$\dot{x}_1 = x_2, \quad \dot{x}_2 = -\frac{g}{L} \sin(x_1)$$

Why nonlinear? The $\sin(x_1)$ term cannot be solved by matrix exponentials. Any dictionary without $\sin \theta$ will fail to represent the dynamics exactly.

Discrete simulation via Euler integration:

$$x_{k+1} = x_k + \Delta t f(x_k)$$

Key question

Which functions go into the dictionary $\Psi(x)$?

- 1 **Closure** — $\frac{d}{dt}\psi_i$ stays inside the dictionary
- 2 **Physics / EOM insight** — include all nonlinear terms from the equations of motion
- 3 **Eigenfunction property** — eigenfunctions diagonalise K
- 4 **Richness & non-redundancy** — well-conditioned SVD
- 5 **State reconstruction** — always include x itself

Criterion 1 — Closure

Closure property

The dictionary is closed if $\frac{d}{dt}\psi_i(x)$ can be written as a linear combination of other dictionary functions.

Without closure: No finite matrix K can satisfy $K\Psi(x_k) = \Psi(x_{k+1})$ exactly. More data does not fix this.

Example — pendulum with only $\{\theta, \dot{\theta}\}$:

$$K \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \dot{\theta} \\ -\frac{g}{L} \sin \theta \end{bmatrix}$$

$\sin \theta$ is not in $\{\theta, \dot{\theta}\} \Rightarrow$ no 2×2 matrix K satisfies this.

Closure — Chain of Forced Terms

Observable	$\frac{d}{dt}\psi_i$	New term?
θ	$\dot{\theta}$	add $\dot{\theta}$
$\dot{\theta}$	$-\frac{g}{L} \sin \theta$	add $\sin \theta$
$\sin \theta$	$\cos \theta \cdot \dot{\theta}$	add $\cos \theta$
$\cos \theta$	$-\sin \theta \cdot \dot{\theta}$	already in ✓
$\dot{\theta}^2$	$-2\frac{g}{L}\dot{\theta} \sin \theta$	already in ✓
$\mathcal{H}$	0	zero leakage

Final dictionary: $\Psi(x) = [\theta, \dot{\theta}, \sin \theta, \cos \theta, \dot{\theta}^2, \mathcal{H}]^\top \in \mathbb{R}^6$

Criterion 3 — Eigenfunctions and Why They Matter

Eigenfunction equation

φ is a Koopman eigenfunction with eigenvalue λ if:

$$\nabla \varphi(x)^\top f(x) = \lambda \varphi(x)$$

Along any trajectory: $\varphi(x(t)) = e^{\lambda t} \varphi(x_0)$

Why eigenfunctions?

- Ordinary closed dict $\Rightarrow K$ is dense (all modes coupled)
- Eigenfunction dict $\Rightarrow K = \text{diag}(\lambda_1, \dots, \lambda_N)$
- Each mode evolves independently as $e^{\lambda_i t}$

The Hamiltonian as a Koopman Eigenfunction

Pendulum energy:

$$\mathcal{H} = \frac{1}{2}mL^2\dot{\theta}^2 + mgL(1 - \cos \theta)$$

Verification (substitute $\ddot{\theta} = -\frac{g}{L} \sin \theta$):

$$\nabla \mathcal{H}^\top f(x) = \dot{\theta} \cdot mgL \sin \theta - \frac{g}{L} \sin \theta \cdot mL^2 \dot{\theta} = 0$$

$$\Rightarrow \lambda = 0$$

Significance:

- $\lambda = 0$: energy is conserved, zero leakage

After Observable Selection — The Pipeline

- 1 **Lift** every snapshot: $x_k \mapsto \Psi(x_k) \in \mathbb{R}^N$
- 2 **Build** snapshot matrices X, Y
- 3 **Solve** $K = YX^\dagger$ via least squares + SVD
- 4 **Eigendecompose** K : extract λ_i, w_i
- 5 **Recover** eigenfunctions $\varphi_i(x) = w_i^\top \Psi(x)$ and modes ϕ_i
- 6 **Reconstruct / predict** using spectral formula
- 7 **Validate** and iterate if needed

Step 1 — Lifting

Input: m state snapshots $x_1, \dots, x_m \in \mathbb{R}^2$ (pendulum).

Lifting:

$$\Psi(x_k) = \begin{bmatrix} \theta_k \\ \dot{\theta}_k \\ \sin \theta_k \\ \cos \theta_k \\ \dot{\theta}_k^2 \end{bmatrix} \in \mathbb{R}^5$$

Geometric meaning: The 2D nonlinear trajectory becomes a curve in $\mathbb{R}^5$. If the dictionary is closed, this curve lies on a **linear subspace** — the nonlinear geometry is “straightened out”.

Step 2 — Snapshot Matrices

Arrange lifted snapshots into two time-shifted matrices:

$$X = [\Psi(x_1) \ \Psi(x_2) \ \cdots \ \Psi(x_{m-1})] \in \mathbb{R}^{N \times (m-1)}$$

$$Y = [\Psi(x_2) \ \Psi(x_3) \ \cdots \ \Psi(x_m)] \in \mathbb{R}^{N \times (m-1)}$$

Column k of X and column k of Y are a **matched pair** — same trajectory, one step apart.

Koopman hypothesis:

$$\exists K \in \mathbb{R}^{N \times N} : \quad Y = KX$$

The same linear map K propagates every lifted state forward by one step.

Step 3 — Solving for K via Least Squares and SVD

Minimise prediction error:

$$K = \arg \min_{\tilde{K}} \|\tilde{K}X - Y\|_F^2 \Rightarrow \boxed{K = YX^\dagger}$$

Compute pseudoinverse via SVD: $X = U\Sigma V^\top$

$$X^\dagger = V\Sigma^{-1}U^\top \Rightarrow K = YV\Sigma^{-1}U^\top$$

Singular values σ_i tell you:

- Large σ_i : observable direction well-represented — keep
- Near-zero σ_i : redundant observable — truncate

Step 4 — Eigendecomposition of K

Solve:

$$KW = W\Lambda, \quad \Lambda = \text{diag}(\lambda_1, \dots, \lambda_r)$$

Eigenvalue interpretation (discrete time):

- $|\lambda_i| < 1$: mode decays (stable)
- $|\lambda_i| > 1$: mode grows (unstable)
- $|\lambda_i| = 1, \text{Im}(\lambda_i) \neq 0$: oscillatory
- $\lambda_i = 1$ exactly: conserved quantity ($\mathcal{H}$)

Eigenvector w_i : Shows how mode i is distributed across the dictionary. The matrix W is the change-of-basis from observable coordinates to eigenfunction coordinates — where K is diagonal.

Step 5 — Eigenfunctions and Modes

Koopman eigenfunction (scalar field over state space):

$$\varphi_i(x) = w_i^\top \Psi(x)$$

Verification: $\varphi_i(F(x)) \approx \lambda_i \varphi_i(x)$ ✓

Level sets $\{\varphi_i(x) = c\}$ are invariant manifolds.

For $\lambda_i = 0$: level sets are *fixed* — the closed orbits in the phase portrait.

Koopman modes (amplitude shape of mode i):

$$\Phi = YV_r \Sigma_r^{-1} W, \quad \phi_i = \Phi_{:,i} \in \mathbb{R}^N$$

ϕ_i gives the contribution of mode i to each observable (θ , $\dot{\theta}$, etc.).

Step 6 — Reconstruction and Prediction

Initial amplitudes: $b_i = \varphi_i(x_0) = w_i^\top \Psi(x_0)$

Spectral reconstruction:

$$g(x(t)) = \sum_{i=1}^r b_i \phi_i^{(g)} e^{\lambda_i t}$$

Each term is a pure exponential — no coupling between modes.

Forward prediction in lifted space:

$$\Psi(x_{k+j}) \approx K^j \Psi(x_k) = W \Lambda^j W^{-1} \Psi(x_k)$$

Raises each eigenvalue to the j -th power — $O(r)$ operations.

This is the **Koopman spectral decomposition** — the nonlinear analogue of a Fourier series.

Four checks after computing K :

- 1 **Reconstruction error:** $\varepsilon = \|Y - KX\|_F / \|Y\|_F \ll 1$
- 2 **Eigenvalue location:** for a conservative system, all $|\lambda_i| = 1$ (unit circle). Spiral $\Rightarrow$ bad dictionary.
- 3 **Eigenfunction smoothness:** plot $\varphi_i(x)$ over a state-space grid. Noisy/discontinuous $\Rightarrow$ bad dictionary.
- 4 **Prediction horizon:** compare $K^j \Psi(x_0)$ to the true trajectory. Early divergence $\Rightarrow$ enrich dictionary, repeat from Step 1.

Complete EDMD Pipeline

Step	Operation	Output
1	Lift: $\Psi(x_k)$	Trajectory in $\mathbb{R}^N$
2	Build X, Y	Matched snapshot matrices
3	$K = YX^\dagger$ via SVD	Koopman matrix K
4	$KW = W\Lambda$	Eigenvalues, eigenvectors
5a	$\varphi_i(x) = w_i^\top \Psi(x)$	Eigenfunctions
5b	$\Phi = YV\Sigma^{-1}W$	Koopman modes
6	$g(t) = \sum b_i \phi_i e^{\lambda_i t}$	Prediction
7	Check $\varepsilon, \lambda_i , \varphi_i$	Validate / iterate

Dynamic Mode Decomposition (DMD)

Special case: observable = state itself, $\Psi(x) = x$

$$x_{k+1} = Ax_k, \quad A = YX^\dagger$$

	DMD	EDMD
Observable	$\Psi(x) = x$	$\Psi(x) \in \mathbb{R}^N, N \gg n$
Nonlinearity	Poorly handled	Well handled
Closure	Rarely exact	Can be exact
Error type	Structural (irreducible)	Fitting (reducible)

Use DMD as a first pass; follow with EDMD for nonlinear systems.

EDMD Matrix Formulation

Define empirical Gram matrices:

$$G = \frac{1}{M} \sum_{k=1}^M \Psi(x_k) \Psi(x_k)^\top, \quad A = \frac{1}{M} \sum_{k=1}^M \Psi(x_k) \Psi(x_{k+1})^\top$$

EDMD solution:

$$K = G^{-1}A$$

- G : covariance of observables over data
- A : cross-covariance between consecutive steps
- $K = G^{-1}A$: regression of future observables on current ones

Koopman Eigenvalues and Eigenfunctions

Eigenvalues λ_i : fundamental temporal dynamics.

- $\text{Re}(\lambda_i) < 0$: decay $\text{Im}(\lambda_i) \neq 0$: oscillation $\lambda_i = 0$: conserved

Eigenfunctions $\varphi_i(x) = \xi_i^\top \Psi(x)$:

- Scalar fields over state space
- Level sets = invariant manifolds of the dynamics
- Coordinates in which K is diagonal — system looks linear

JUICE context:

- Eigenvalues $\leftrightarrow$ tidal forcing frequencies
- Extra eigenvalues $\leftrightarrow$ ocean–ice coupling modes

Spectral Interpretation

Nonlinear dynamics expressed as a linear combination of exponential modes:

$$g(x_k) = \sum_i \varphi_i(x_0) e^{\lambda_i k}$$

	Fourier	Koopman
Basis	$e^{j\omega t}$	$e^{\lambda_i t}$
Decomposition	Frequency spectrum	Eigenvalue spectrum
Coordinates	Frequency domain	Eigenfunction coordinates
Exact when	Always (linear)	Dictionary is closed

Conclusion

- Koopman operator: **exact linear representation** of nonlinear systems in function space.
- Observable selection: closure $\rightarrow$ physics insight $\rightarrow$ eigenfunction property $\rightarrow$ richness $\rightarrow$ state reconstruction.
- **Closure** is structural — without it, no finite K is correct regardless of data.
- **Pipeline**: lift $\rightarrow$ snapshot matrices $\rightarrow K = YX^\dagger$ via SVD $\rightarrow$ eigendecompose $\rightarrow$ eigenfunctions/modes $\rightarrow$ validate.
- **Eigenfunctions** diagonalise K — independent exponential modes (nonlinear Fourier).
- **Hamiltonian** ($\lambda = 0$): guaranteed by physics, reduces multivariable control to a scalar problem.

Matlab Simulations of DMD

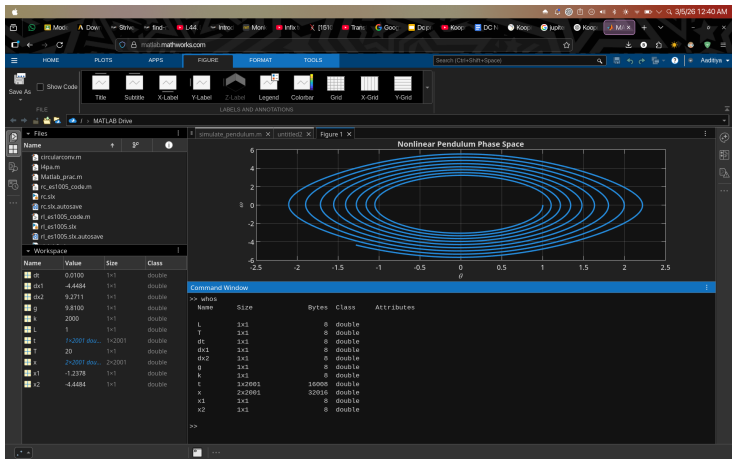


Figure: Nonlinear pendulum phase space

Matlab Simulations of DMD

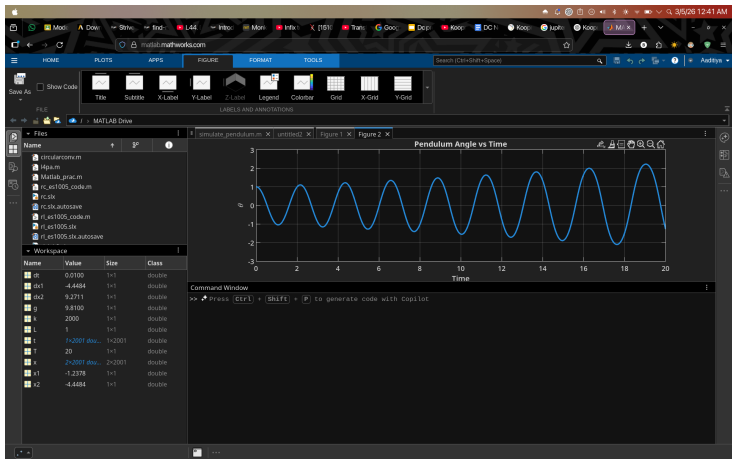


Figure: Pendulum angle vs time

Matlab Simulations of DMD

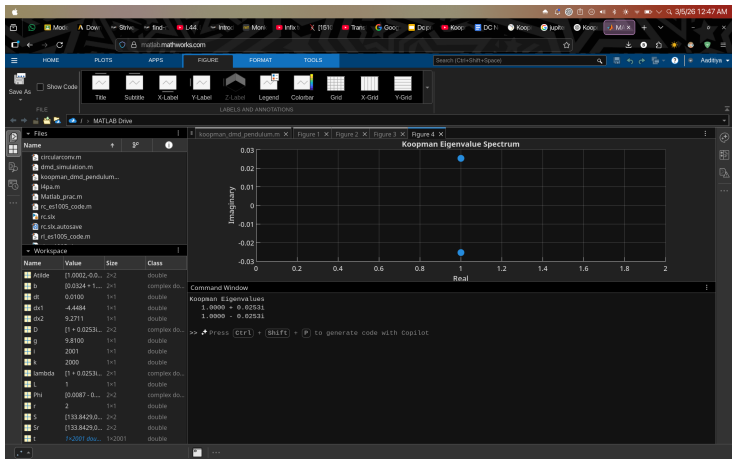


Figure: Koopman eigenvalue spectrum (DMD)

Matlab Simulations of DMD



Figure: True vs DMD reconstruction

Matlab Simulations of Extended DMD

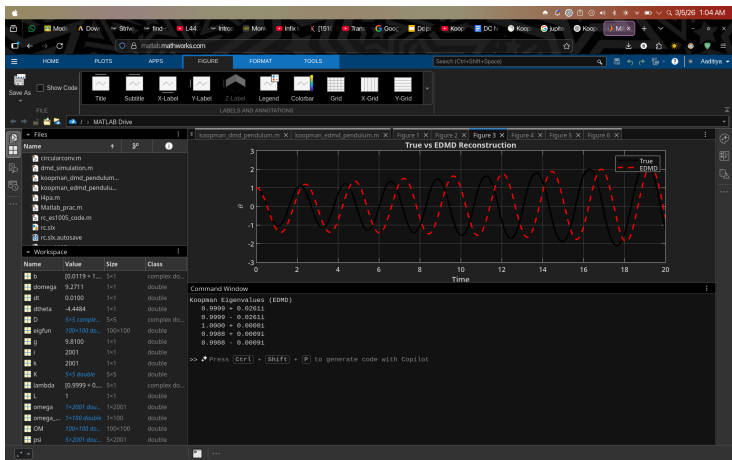


Figure: True vs EDMD reconstruction

Matlab Simulations of Extended DMD

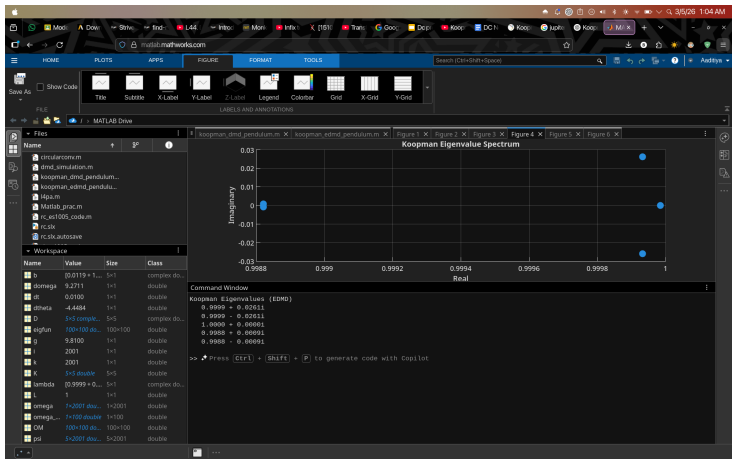


Figure: Koopman eigenvalue spectrum (EDMD)

Matlab Simulations of Extended DMD



Figure: Example Koopman eigenfunction

Matlab Simulations of Extended DMD

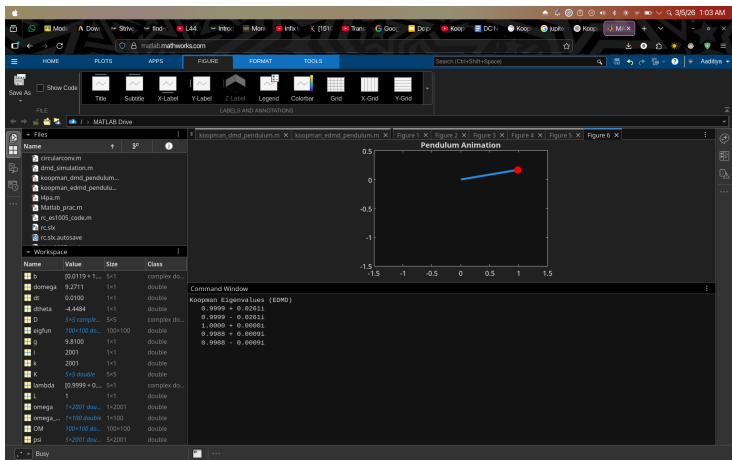


Figure: Pendulum simulation animation

Koopman operator theory provides a powerful framework for nonlinear systems.

- Converts nonlinear dynamics to linear evolution in function space
- Observable selection via closure, physics, and eigenfunction criteria
- Full pipeline: lift $\rightarrow$ SVD $\rightarrow$ eigendecompose $\rightarrow$ predict
- Global spectral analysis — valid far from equilibrium
- Applications: control (KRONIC/SDRE), fluid dynamics, robotics, planetary science (JUICE)

One scalar (Hamiltonian) replaces a full matrix optimal control problem.